

Guwahati Urban Development Prediction Platform — Architecture & Build Plan

Sep 26, 2026 · @Biplab Das

A 10-year (2026–2036) predictive urban-development platform for Guwahati, combining satellite change-detection, government-policy signal extraction, and hybrid ML forecasting, rendered on an Esri World Imagery basemap and built toward a multi-city SaaS product.

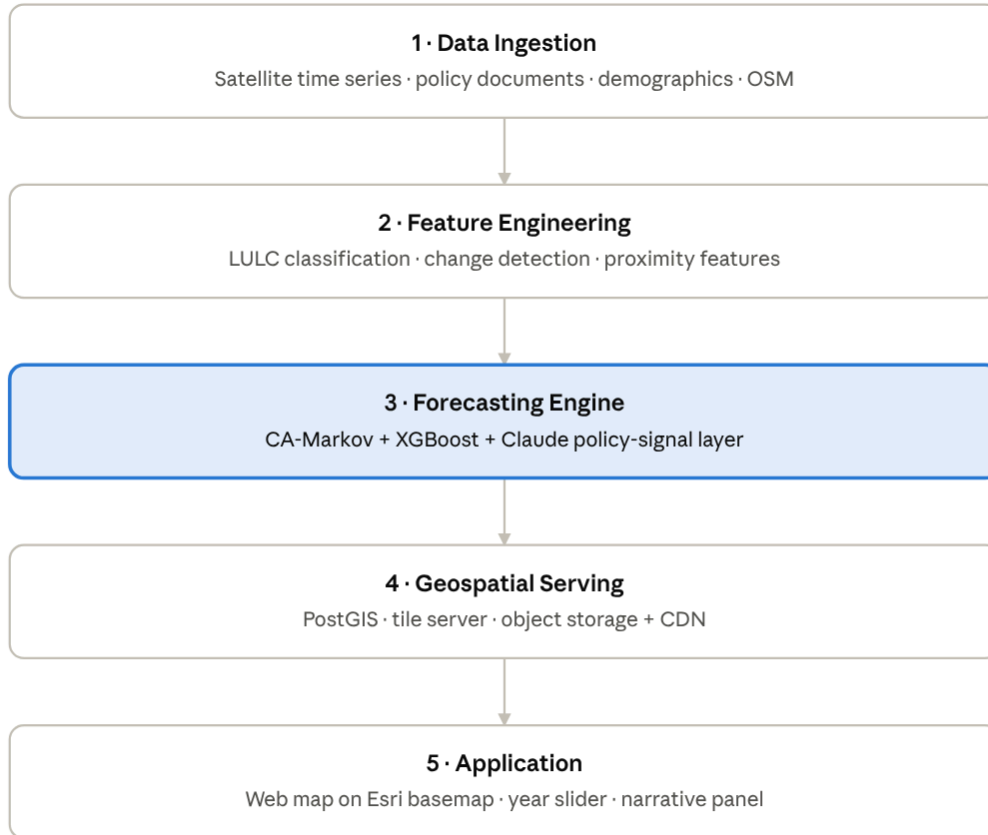
1. System overview

Five layers, each independently replaceable: **ingestion** (satellite time series, government policy documents, demographic proxies, OSM), **feature engineering** (land-use classification, change detection, proximity features), **forecasting** (hybrid CA-Markov + gradient-boosted trees + an LLM policy-signal layer), **geospatial serving** (PostGIS, tile server, CDN), and **application** (web map on the Esri World Imagery basemap, year slider, narrative panel).

The forecasting layer is deliberately hybrid, not a single model: cellular-automata/Markov transition modelling handles the physical land-use dynamics (the same family as SLEUTH and Land Change Modeler), gradient-boosted trees weight *where* growth is more likely given proximity to announced infrastructure, and Claude reads government master plans and budget notifications to convert policy announcements into structured spatial priors that feed the other two.

A macro layer sits underneath all of this: Assam GSDP growth forecasts and India GDP growth forecasts act as a scaling factor on the whole forecasting engine — a high-growth national/state trajectory pulls the CA-Markov and XGBoost outputs toward faster built-up expansion, a slowdown pulls them back. This is separate from the VIIRS night-lights proxy, which only measures historical ground activity, not the forward-looking growth rate a 10-year forecast needs.

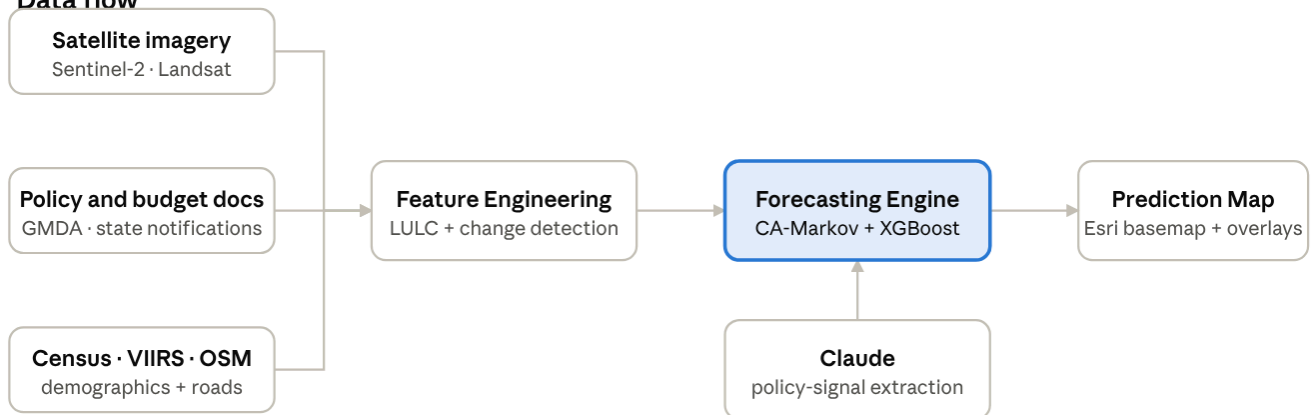
Five-layer architecture



component architecture · 5 layers, ingestion to application

The forecasting layer (highlighted) is the only layer that combines three distinct techniques; every other layer is a single, swappable component.

Data flow



data flow · three input streams converge into forecasting

The Forecasting Engine is the only box with two independent inputs: the numeric feature stream from the left, and Claude's structured policy-signal extraction from below. Losing either input degrades the forecast quality but doesn't break the pipeline.

4. Requirements

Category	Item	Notes
Imagery (analysis)	Sentinel-2, Landsat via Google Earth Engine	Free for research use; 2015–present (Sentinel-2), 1972–present (Landsat)
Imagery (basemap)	Esri ArcGIS Location Platform	2M free basemap tiles/month, then \$0.15/1,000; commercial deployment license included in the developer subscription
Policy data	GMDA master plan, Assam govt notifications, NE region scheme documents	Manually sourced; this is the highest-leverage, least automatable input
Demographic data	Census of India, VIIRS night-lights, WorldPop/LandScan, OSM	2027 Census house-listing began April 2026; population count reference date is 1 March 2027 — falls inside the 10-year window, so architect for a mid-window ground-truth refresh
Compute	GPU only for periodic LULC classifier training	Spot/preemptible instances, or stay inside Earth Engine's own compute
Database	PostGIS	Vector storage for predicted built-up polygons, infra overlays, demographic grid
Tile serving	Martin or TiTiler + object storage + CDN	Either cloud works equally here
Dev tooling	Claude Code in VS Code	Covers ETL scripts, API, frontend end-to-end per your stated workflow
Cloud	GCP for the analytics layer (Earth Engine advantage); AWS or GCP for API/DB/frontend	See SaaS section for the licensing note on redistribution

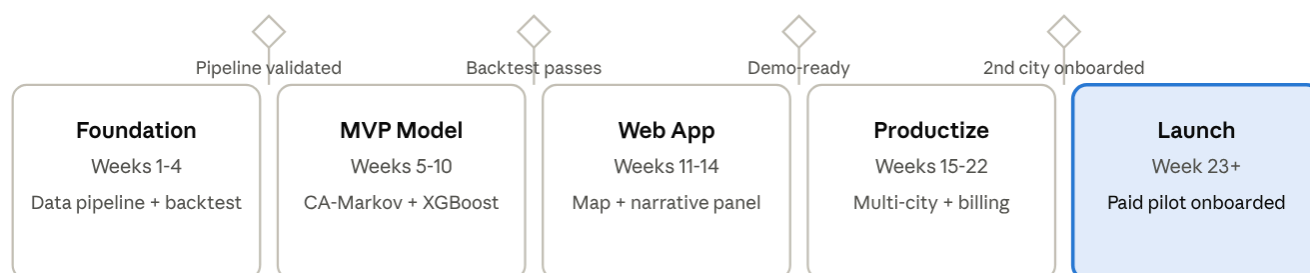
Category	Item	Notes
Economic growth scenarios	Assam GSDP growth forecasts (Finance Dept budget, NITI Aayog) + India GDP growth forecasts (RBI MPC, IMF, Economic Survey)	Forward-looking scaling factor for the whole model — VIIRS/night-lights only give historical ground truth, not the growth trajectory a 10-year forecast needs

5. SaaS path

Three changes take this from a Guwahati demo to a sellable product:

1. **Generalize the pipeline** — parameterize by city (bounding box, local planning authority, local infra dataset) so onboarding a second city is a config change, not a rewrite.
2. **Multi-tenancy and billing** — auth (Cognito/Auth0/Clerk), Stripe subscriptions, per-tenant usage metering for the Esri tile and Claude API costs you're passing through.
3. **Licensing check before commercial launch** — the ArcGIS Location Platform developer subscription includes a commercial deployment license for displaying the basemap in a product you sell. Confirm current terms before launch; this covers your app *displaying* Esri's basemap, not reselling their imagery as your own data product. Native Maxar tasking (beyond the free World Imagery mosaic) is a separate, far more expensive relationship directly with Maxar — skip it for the MVP and any early SaaS tiers.

Phased build plan



roadmap · 5 phases, 4 gates, foundation to commercial launch

Timeline assumes one person working part-time alongside your other consulting work; a dedicated full-time effort could compress this to roughly 14-16 weeks total. Each gate is a go/no-go checkpoint, not a soft milestone — the backtest gate in particular should block moving to the web app phase if 2011→2021 accuracy is poor, since a bad model is worse than no model in a paid product.

6. Budget (lowest-of-lowest MVP)

Item	Cost	Notes
Earth Engine (Sentinel/Landsat analysis)	Free	Non-commercial/research use
Esri ArcGIS Location Platform (basemap)	Free	2M tiles/month is far more than an MVP needs
OSM, VIIRS, Census/govt data	Free	
Compute (small VM for API + PostGIS, or free-tier for training)	\$0-30/month	
Object storage + CDN for tiles	\$5-10/month	
Domain + static frontend hosting	\$0-10/month	
Claude API (policy-synthesis layer, MVP volume)	\$10-20/month	
Total	~\$30-70/month	Plus your own time

Budget jumps into the low hundreds/month once you commercialize with Esri's paid tier at scale, add Maxar tasking, or retrain the model regularly — none of which is needed to prove the concept.

7. Open questions and risks

- ☐ Confirm current ArcGIS Location Platform commercial-deployment terms before any paid launch
- ☐ Decide whether to wait for the 2027 Census (reference date 1 March 2027) before finalizing the demographic module, or ship with VIIRS/WorldPop proxies and backfill later
- ☐ Line up who reads and structures the GMDA/state policy documents — this is manual, ongoing, and the hardest part to automate
- ☐ Guwahati sits on the Brahmaputra flood plain — flood-risk zoning should be a hard constraint on the growth model, not just a covariate
- ☐ A single-city backtest (train on 2011-2016 imagery, validate against 2016-2021 actuals) is the real test of whether the hybrid model is worth productizing before any SaaS investment

8. Competitive landscape

No existing India-based solution combines a forward 10-year forecast, an LLM policy-signal layer, and multi-city SaaS delivery in one product. What exists sits in three tiers, each useful as an input or benchmark rather than a competitor:

Tier	Example	What it is	Gap vs. this platform
Government (retrospective)	ISRO Bhuvan — Urban Growth Monitoring	Multi-temporal, multi-spectral change-detection viewer covering many Indian cities since the 1990s	Historical viewer only, no forward forecast — worth pulling in as a free ingestion source
Government (infra data)	PM Gati Shakti National Master Plan	GIS platform integrating 1,400+ infrastructure data layers across 44+ ministries and all states	The best available source for the "government impetus" signal — not predictive itself
Commercial	SatSure (Bengaluru)	Earth-observation analytics tailored to Indian conditions, spanning climate, agriculture, urban expansion and infrastructure	Broader EO/analytics platform, not a packaged 10-year urban-growth-prediction SaaS
Commercial (PoC)	Elementr.io	Modular demonstrator fusing satellite imagery, vector layers and socioeconomic data to predict urban growth and gentrification	Proof-of-concept stage, not a shipped product
Academic	Hybrid ML/XAI studies on Lucknow and Hyderabad	Published models combining Landsat imagery, socioeconomic data and ensemble ML (XGBoost, Random Forest, LSTM) to forecast urban expansion	Validates the forecasting approach, but research papers, not deployed software

Sources

- [ISRO Bhuvan — Urban Growth Monitoring](#)
- [PM Gati Shakti and GIS — Esri India](#)

- [SatSure Analytics — Earth observation grant](#)
- [Elementr — Geo AI Solutions](#)
- [Hybrid intelligence and explainable AI for urban growth prediction — Lucknow](#)